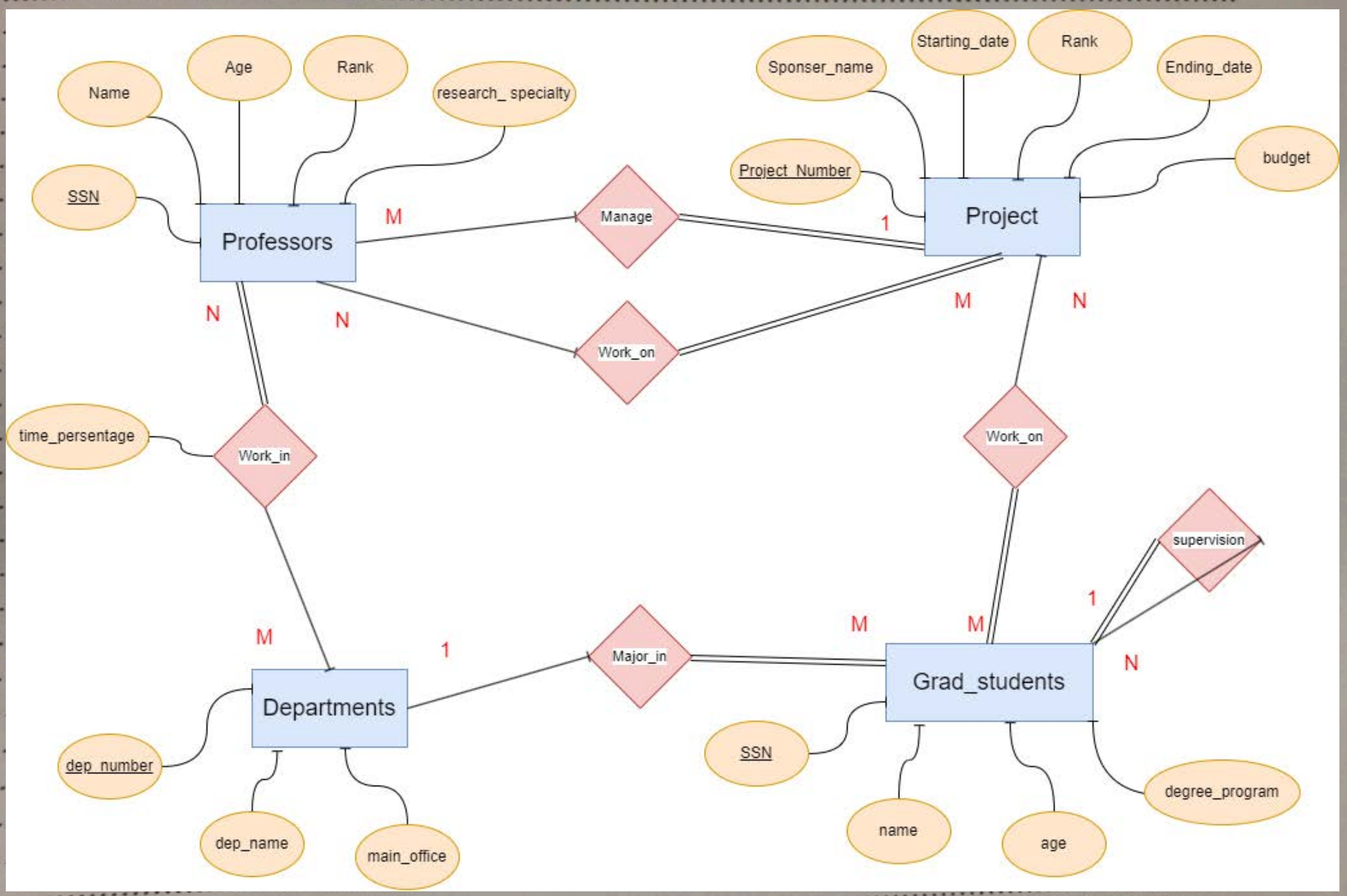
Mid-Term Exam (15 Points)

الاسم: ..... رقم السكشن: .....

Q1) Consider the following set of requirements for a UNIVERSITY database: **(7 Points)**

- Professors have an SSN, a name, an age, a rank, and a research specialty.
- Projects have a project number, a sponsor name (e.g., NSF), a starting date, an ending date, and a budget.
- Graduate students have an SSN, a name, an age, and a degree program (e.g., M.Sc. or Ph.D.).
- Departments have a department number, a department name, and a main office.
- Each project is managed by one professor (known as the project's principal investigator).
- Each project is worked on by one or more professors (known as the project's co- investigators). Professors can manage and/or work on multiple projects.
- Each project is worked on by one or more graduate students (known as the project's research assistants). Graduate students can work on multiple projects.
- Professor's work in one or more departments and for each department that they work in, a time percentage is associated with their job.
- Graduate students have one major department in which they are working on their degree. • Each graduate student has another, more senior graduate student (known as a student advisor) who advises him or her on what courses to take.

Design an ER Diagram to and specify key attributes of each entity type and structural constraints on each relationship type.

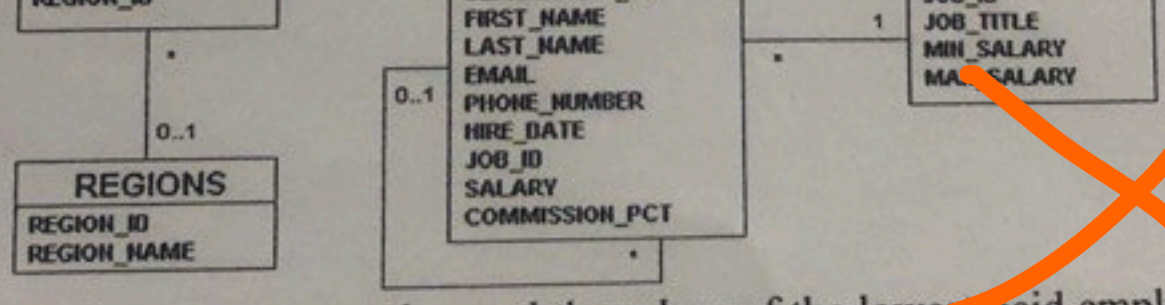


Subject: IS200 Database
Time allowed: 60 Minute
Exam (15 Points)

Q2) For Each Question, Choose the Correct Answer from the Multiple-Choice List: **(5 Points)**

1. For each attribute of a relation, there is a set of permitted values, called the _____ of that attribute.
   A) Schema B) Relation
   C) Set D) Domain
2. Which one of the following uniquely identifies the elements in the relation?
   A) Composite key B) Secondary Key
   C) Primary key D) Foreign key
3. A _____ indicates an absent value that may exist but be unknown or that may not exist at all.
   A) Old value B) New value
   C) Null value D) Empty tuple
4. A _____ constraint requires that an entity belong to no more than one lower-level entity set.
   A) Overlap B) Disjointness
   C) Relational D) Uniqueness
5. There are similarities between the instructor entity set and the secretary entity set in the sense that they have several attributes that are conceptually the same across the two entity sets: namely, the identifier, name, and salary attributes. This process is
   A) Similarity B) Specialization
   C) Generalization D) Commonality
6. All aggregate functions except _____ ignore null values in their input collection.
   A) Count(attribute) B) Count(*)
   C) Sum D) Avg
7. SELECT * FROM employee
   What type of statement is this?
   A) DCL B) DML
   C) DDL D) DVL
8. The entity set person is classified as student and employee. This process is called _____
   A) Similarity B) Generalization
   C) Specialization D) Inheritance
9. The attribute name could be structured as an attribute consisting of first name, middle initial, and last name. This type of attribute is called
   A) Composite B) Multivalued
   C) Derived D) Simple
10. The primary key must be
    A) Not null
    B) Unique
    C) Both Unique and Not null
    D) Either Unique or Not null

Q3) Consider the following schema, write the equivalent SQL statement for each report: **(3 Points)**



Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is $6,000 or less. Sort the output in descending order of salary.

we will need functions to solve this and it's not in our curriculum for midterm